

Real-Time Detection and Estimation of Denial of Service Attack in Connected Vehicle Systems

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Abstract—Advanced connectivity features in today's smart vehicles are giving rise to several promising intelligent transportation technologies. Connected vehicle system is one among such technologies, where a set of vehicles can communicate with each other and the infrastructure via communication networks. Connected vehicles have the potential to improve the traffic throughput, minimize the risk of accidents and reduce vehicle energy consumption. Despite these promising features, connected vehicles suffer from the safety and security issues. Especially, vehicle-to-vehicle and vehicle-to-infrastructure communication make the connected vehicles vulnerable to cyber attacks. In order to improve safety and security, advanced vehicular control systems must be designed to be resilient to such cyber attacks. The first step of designing such attack-resilient control system is detection of the occurrence of the cyber attack. In this paper, we address this need and propose a real-time scheme that can potentially 1) detect the occurrence of a particular cyber attack, namely denial of service; and 2) estimate the effect of the attack on the connected vehicle system. The scheme consists of a set of observers, which are designed using sliding mode and adaptive estimation theory. The mathematical convergence properties of the observers are analyzed via Lyapunov's stability theory. Finally, simulation demonstrates the performance of the approach and the robustness of the scheme under several forms of uncertainties.

Index Terms—Connected vehicles, denial of service attack, detection, estimation.

I. INTRODUCTION

INTELLIGENT Transportation Systems (ITS) is one of the potential solutions that addresses the issues of traffic congestion, safety and emission. Connected vehicle system is one of such ITS which can potentially minimize the risk of accidents and increasing traffic throughput. However, connected vehicular networks are vulnerable to packet dropping [1], [2], [3], communication induced-delay, [4], [5], scheduling issues and malicious cyber attacks [6], [7]. Hence, the vehicular control systems must be capable of handling

and surviving such adverse situations. In line with these requirements, we propose a diagnostic scheme for vehicular control system that can detect the occurrence of a particular cyber attack.

There is a whole body of work in the existing literature investigating various modelling and control schemes for connected vehicles. Three of the most common and well-explored models of connected vehicles are (i) Cooperative Cruise Control (CCC) [8] [9] [10], and (ii) Cooperative Adaptive Cruise Control (CACC) [11] [12], and (iii) car following model [13]. In this paper we mainly focus on the CACC, which is more realistic in term of using communication network among connected vehicles. Another important aspect of the connected vehicles is the string stability which was discussed in [14] and [15]. However, communication networks such as DSRC is subjected to unintended network failures e.g. latency and packet drop out [16], [5] as well as intended cyber-attacks. Many existing literature evaluates the performance of DSRC considering packet delivery ratio and network induced delay [17], [18]. Hence, analyzing the performance of the connected vehicles in the presence of the aforementioned network failures is crucial [8] [9].

Stability of connected vehicles equipped with CACC algorithm is discussed in [19] and [20] under normal delay in communication network. Oncu et.al also consider actuator delay as well as communication delay due to limited bandwidth in a platoon of connected vehicles to analyze the string stability [19]. Moreover, Kazerooni and Ploeg discussed the string stability when the inter spacing policy between vehicles is dynamic and malfunctioning of redundant sensing/communication data occurs [21]. To partially maintain the desired string stability properties of CACC, Ploeg et.al develop a control strategy for graceful degradation of CACC based upon the preceding vehicle's acceleration estimation using on-board sensors [22]. To assure the performance, safety and security of the platoon of connected vehicles equipped with CACC, the string should be stable and attenuating the effect of disturbances along the vehicle string. The string stability of CACC-equipped vehicle platoon is discussed and analyzed via L_2 , L_∞ [23] and L_P [15] stability analysis methods under unreliable communication network. The above review suggests that modeling, control and stability analysis of connected vehicles are well studied in existing literature. However, the problem of real-time detection of cyber attacks on connected vehicles is significantly under-explored despite being extremely critical.

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Cyber attacks calls for significant attention due to the its severe consequences of threatening human lives and safety. However, very few study exist on cyber security of connected vehicles and more specifically cooperative adaptive cruise control. Dadras *et.al.* in [24], explore the capability of attackers to intrude vehicles and show that, the attacker is theoretically capable of taking control over the individual position and velocity of vehicles in the platoon. Moreover, few studies investigate the effect of network failures e.g. packet dropping [25], [26] and cyber-attacks e.g. DoS [27] on connected vehicles. Lei *et.al* discuss an infinity repeated game for cooperative authentication in a platoon of connected vehicles equipped with CACC as a way to recognize false identities and messages as well as saving resources [28]. Moreover, in [29] cyber-attacks on a platoon of connected vehicles is discussed and a safety-security co-design engineering process is proposed to develop functional security requirements for a safe platoon. Stability of string under data jamming attack is investigated in [30]. Furthermore, Amoonzadeh *et.al* study impacts of cyber attacks and sensor tampering in a platoon of vehicles equipped to CACC strategy. Their study illustrates that insider attacks can lead to instability of the string and degrading the CACC strategy to ACC can be a potential countermeasure to cyber attacks [31]. However, in our paper, we address this issue and develop a cyber attack detection scheme. In this paper, we develop a more essential countermeasure to cyber attack with sufficient information of the effect of the attack to avoid losing benefits of CACC algorithm.

Several forms of cyber attacks can be possible in connected vehicle system such as: i) *Denial of Service (DOS)*: The attacker keeps the network busy with fake requests and consequently, the network is unavailable to process the legitimate requests [32], [33]; ii) *False Data Injection*: the attacker tries to affect the integrity of the transmitted packets by modifying their payloads [34], [35]; iii) *Replay Attack*: The attacker intercepts data of the system and re-transmits it while degrading the system functionality [36]. In this work, we focus on DoS attack which is one of the most common attacks in communication networks and Cyber Physical Systems (CPS). Very few literature explore the DoS in CPS from a control-oriented viewpoint [37], [38]. However, none of these approaches are designed for connected vehicle systems. In this paper, we attempt to address this research gap regarding the DoS attack detection and estimation in connected vehicles. The main contribution of the present paper is a control-oriented diagnostic framework for connected vehicle systems that is capable of (i) modeling DoS attack with unknown time-delay; (ii) detecting the occurrence of DoS attack and, (iii) providing an estimate of the effect of the attack. Note that, the estimate of the effect of the attack can be extremely useful for designing secure control system for the vehicles. In this paper we model the DoS attack as a time delay in information processed by the network and design a detection scheme based on adaptive estimation and sliding mode theory. Essentially, the goal of the scheme is to track the delay in the information processing by the DSRC. A DoS occurrence is detected when the delay exceeds a pre-defined threshold.

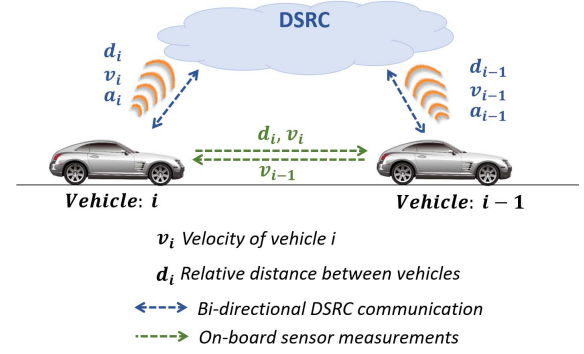


Fig. 1. Platoon of connected vehicles equipped with CACC.

The remainder of this paper is organized as follows. Section II provides a control oriented mathematical modeling of the connected vehicles and DoS attack. Section III describes the diagnostic scheme. Simulation results are discussed in section IV to evaluate the effectiveness of this approach; and finally, Section V offers the concluding remarks of this work.

II. MATHEMATICAL MODELING

In this section, we model the connected vehicles with a linear state space representation of the CACC strategy. Unlike the other existing strategies on modeling the connected vehicles e.g. Cooperative Cruise Control (CCC), the CACC strategy considers the connectivity among the vehicles as a part of the system model. Indeed, an additional information of the acceleration of the preceding vehicle is utilized into the control strategy to reduce the relative distance of the vehicles into the platoon comparing to the CCC strategy.

A. Modeling of Platoon

Consider a homogeneous platoon of connected vehicles equipped with CACC strategy following their leader in a single lane (see Fig.1). Each vehicle in the platoon is assumed to be equipped with on-board sensors to measure the relative distance and velocity with respect to its preceding and the following vehicle. In addition, each vehicle receives the acceleration information of the preceding vehicle through Dedicated Short Range Communication (DSRC) network.

Remark 1: In this study, we assume the leader vehicle follows a dynamic velocity profile. Therefore, the proposed approach is not restricted to constant velocity assumption.

Dynamic of vehicle i in the platoon can be written as:

$$\begin{bmatrix} \dot{d}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} v_{i-1} - v_i \\ a_i \end{bmatrix}, i \in S_m \setminus \{1\} \quad (1)$$

where $S_m = \{i \in N | 1 \leq i \leq m\}$ is the set of all vehicles in the platoon of length m , $d_i = q_{i-1} - q_i + L_i + d_s$ is the distance between vehicle i and $i-1$, q_i and q_{i-1} are the rear bumper position of vehicles i and $i-1$, L_i is the length of vehicle i , d_s is the minimum safety distance between two vehicles, v_i and a_i are the velocity and acceleration of vehicle i . Furthermore, a_i acts as the control input to the vehicle [2]. The control strategy regarding the inter-vehicle spacing is:

$$d_{r,i}(t) = h v_i(t), i \in S_m \setminus \{1\} \quad (2)$$

where $d_{r,i}$ is the reference inter-vehicle distance to be maintained and h is the time headway. Without loss of the generality, consider $L_i = 0$ and $d_s = 0.5$ m for simplicity. The main objective of platooning is to regulate the d_i to $d_{r,i}$, so that the regulation error

$$e_i(t) = d_i(t) - d_{r,i}(t) \rightarrow 0 \quad (3)$$

Substituting (2) and $d_i = q_{i-1} - q_i$ in (3), we have:

$$e_i(t) = q_{i-1}(t) - q_i(t) - hv_i(t) \quad (4)$$

Next, we use the following controller to attain the control objective [12]:

$$\dot{a}_i = -\frac{1}{h}a_i + \frac{1}{h}(k_p e_i + k_d \dot{e}_i) + \frac{1}{h}a_{i-1} \quad (5)$$

Considering (2) and (3), (5) can be written as:

$$\dot{a}_i = \frac{k_p}{h}d_i - (k_p + \frac{k_d}{h})v_i - (k_p + \frac{1}{h})a_i + \frac{k_d}{h}v_{i-1} + \frac{1}{h}a_{i-1} \quad (6)$$

where a_{i-1} and v_{i-1} are the desired acceleration and velocity of the preceding vehicle received through DSRC network. The parameters $k_p, k_d > 0$ are controller gains designed such that (i) the inter-vehicle distance is maintained to $d_{r,i}$ and, (ii) a_i is bounded and changes smoothly. As can be inferred from (6), control signal of vehicle i , a_i , depends on states of vehicle i and information received from preceding vehicle $i - 1$. Considering (1) and (6), a new augmented state space representation for vehicle i is shown in (7), where, a_{i-1} and v_{i-1} are two external inputs of the system related to the preceding vehicle $i - 1$.

$$\begin{bmatrix} \dot{d}_i(t) \\ \dot{v}_i(t) \\ \dot{a}_i(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ \frac{k_p}{h} & -(k_p + \frac{k_d}{h}) & (k_p + \frac{1}{h}) \end{bmatrix} \begin{bmatrix} d_i(t) \\ v_i(t) \\ a_i(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ \frac{k_d}{h} \end{bmatrix} v_{i-1}(t) + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{h} \end{bmatrix} a_{i-1}(t) \quad (7)$$

Next, we make the following assumptions:

Assumption 1: We consider a homogeneous platoon of vehicles. Therefore, all vehicles in the platoon have the same parameters.

Assumption 2: Each vehicle in the platoon measures relative distance with respect to preceding vehicle d_i and following vehicle d_{i+1} .

Remark 2: Vehicle i measures the relative velocity via radar and hence can compute the absolute velocity v_{i-1} of vehicle $i - 1$. Vehicle i also receives acceleration a_{i-1} information of vehicle $i - 1$ via DSRC network which is subjected to network failures and cyber-attacks.

B. Modeling of Denial of Service (DoS) Attack

Denial of Service (DoS) attack is a one of the most typical cyber-attacks in communication networks [32]. Exact definition of DoS attack may vary in different applications, however, the impact of the attack remains similar in all possible forms of the attack e.g. network jamming [39], packet

flooding [40], or IP spoofing [41]. In cyber physical systems such as connected vehicles, DoS attack changes the average service time of the communication network by imposing illegitimate requests. Indeed, the DoS attack induces an additional service time which in turn corresponds to additional delay in the transmission network [42]. In the majority of the existing literature, the effect of DoS attack is modeled in two control-oriented perspectives: (i) packet loss [43], [44], and (ii) time delay [45], [33], and [38]. The former method of modeling represents types of DoS attacks in which the attacker attempts to keep the communication network busy for certain time; such that the transmitting packet of a legitimate user have to wait in the queue for a long time. The packet waits in the queue to be transmitted in a proper transmission time slot and in the meantime, the attacker floods too much data on the network to make packet congestion on the network and consequently the packet will loss. However, assuming a stochastic characterization of the DoS attacks such as packet dropping, would be inherently limiting and would fail to capture the malicious and intelligent nature of an attacker [38]. Hence, the later method of DoS attack modeling corresponds to a set of more intelligent types of DoS attacks in which the attacker does not make his actions too obvious to be detected via packet dropping algorithms. In this types of DoS attack, the attacker changes the hubs into the network communication based upon the existing protocol and attacker capabilities of generating fake router/nodes. Indeed, the attacker changes routers and then floods the packets randomly on the network and tries to increase the service time on the communication network [42]. The effect of DoS attack to increase the service time can be modeled by different types of time delays e.g. (i) constant time delay [42], [46]; and (ii) probabilistic time delay, which can be represented by random variables, e.g. Bernoulli [32], [33] or probabilistic approaches with memory, e.g. Markov Model [47], [48].

In this paper, assume the DoS attacker focuses on the endpoint (the following vehicle) and flood the communication network with excessive amount of packets; therefore, the authorized user (vehicle i) can not access to the DSRC network on time and the acceleration data of the leading vehicle $i - 1$ will be delivered to the follower with a delay. Since the capability of the attacker and his underlying strategy are not known; therefore, in this paper, the delay induced by DoS attack is modeled as unknown constant delay. Considering *Remark 1*, dynamics of vehicle i in (7) under DoS attack can be written as

$$\begin{bmatrix} \dot{d}_i(t) \\ \dot{v}_i(t) \\ \dot{a}_i(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ \frac{k_p}{h} & -(k_p + \frac{k_d}{h}) & (k_p + \frac{1}{h}) \end{bmatrix} \begin{bmatrix} d_i(t) \\ v_i(t) \\ a_i(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ \frac{k_d}{h} \end{bmatrix} v_{i-1}(t) + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{h} \end{bmatrix} a_{i-1}(t - \tau) \quad (8)$$

where vehicle i receives the acceleration information of vehicle $i - 1$ with unknown delay, τ , induced by DoS attack.

Assumption 3: The attacker has a limited capability to keep the network busy. Hence, the effect of DoS attack as the

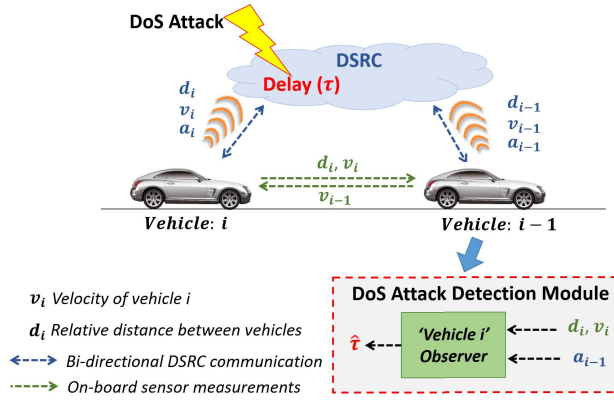


Fig. 2. DoS attack detection and estimation scheme.

unknown delay has an upper bound corresponding to the maximum capability of the attacker. i.e. $\tau \in [0, \tau_{max}]$.

III. REAL-TIME DETECTION AND ESTIMATION SCHEME FOR DoS ATTACK

With the formulation discussed in the previous section, the diagnostic problem is to detect when the delay parameter τ is non-zero and if so, estimate the value of τ . The detection and estimation scheme for DoS attack is presented in Fig. 2.

Remark 3: The DoS detection module is implemented in vehicle $i-1$. As mentioned in the previous section, vehicle $i-1$ has access to the following information: (i) $d_i(t)$ and $v_i(t)$ measured by rear radar of vehicle $i-1$ and, (ii) $a_{i-1}(t)$ directly measured in vehicle $i-1$. Note that, these measurements are not affected by the occurrence of the DoS attack.

The proposed scheme consists of a model-based observer, denoted by *Vehicle i Observer*. Based on the available measurements and the model (8), *Vehicle i Observer* detects the occurrence and estimates size of the delay τ . Before detailing the design of the *Vehicle i Observer*, we make the following assumptions.

Assumption 4: The signal $a_{i-1}(t)$ is at least two times differentiable with respect to time. Furthermore, the derivative is bounded by some finite value, i.e. $|\dot{a}_{i-1}(t)| < \bar{a}_d, \forall t > 0$.

Assumption 5: Using Taylor's series expansion [49], the delayed signal $a_{i-1}(t - \tau)$ can be written as

$$a_{i-1}(t - \tau) = a_{i-1}(t) - \dot{a}_{i-1}(t)\tau + H.O.T \quad (9)$$

where *H.O.T* represents the higher order terms of the Taylor's series expansion. We assume that *H.O.T* is negligible and hence $a_{i-1}(t - \tau) \approx a_{i-1}(t) - \dot{a}_{i-1}(t)\tau$.

A. DoS Attack Detection

Applying *Assumption 5*, the system dynamics (8) can be written as:

$$\dot{d}_i(t) = v_{i-1}(t) - v_i(t), \quad (10)$$

$$\dot{v}_i(t) = a_i(t), \quad (11)$$

$$\begin{aligned} \dot{a}_i(t) = & \frac{k_p}{h}d_i(t) - \left(k_p + \frac{k_d}{h}\right)v_i(t) - \left(k_p + \frac{1}{h}\right)a_i(t) \\ & + \frac{k_d}{h}v_{i-1}(t) + \frac{1}{h}\left(a_{i-1}(t) - \dot{a}_{i-1}(t)\tau\right), \end{aligned} \quad (12)$$

We choose the following structure for *Vehicle i Observer*.

$$\dot{\hat{v}}_i(t) = L_v \text{sgn}\left(v_i(t) - \hat{v}_i(t)\right), \quad (13)$$

$$\begin{aligned} \dot{\hat{a}}_i(t) = & \frac{k_p}{h}d_i(t) - \left(k_p + \frac{k_d}{h}\right)\hat{v}_i(t) - \left(k_p + \frac{1}{h}\right)\eta(t) \\ & + \frac{k_d}{h}v_{i-1}(t) + \frac{1}{h}\left(a_{i-1}(t) - \dot{a}_{i-1}(t)\hat{\tau}\right) \\ & + L_a\left(\eta(t) - \hat{a}_i(t)\right), \end{aligned} \quad (14)$$

$$\dot{\hat{\tau}}(t) = -\frac{L_b}{h}\dot{a}_{i-1}(t)\left(\eta(t) - \hat{a}_i(t)\right), \quad (15)$$

where $\hat{v}_i(t)$, $\hat{a}_i(t)$ are the estimated relative velocity and acceleration, respectively; $\hat{\tau}(t)$ is the estimated delay; L_v, L_a, L_b are the constant observer gains to be designed; $v_{i-1}(t)$ and $a_{i-1}(t)$ are measured via on-board sensors in *Vehicle i-1*; $d_i(t)$ and $v_i(t)$ are measured by *Vehicle i-1* using radar; $\eta(t)$ is a filtered version of the signal $L_v \text{sgn}\left(v_i(t) - \hat{v}_i(t)\right)$. Next, we state the main result of the proposed approach.

Main Result: Consider the system model described in (10)-(12), and the observer structure (13)-(15). If the assumptions 1-5 hold true, then the estimated value of DoS attack, $\hat{\tau}$, converges to its true value τ as $t \rightarrow \infty$, given the the observer gains satisfy the following conditions:

$$L_a, L_b > 0, \quad (16)$$

$$L_v > |a_i(t)| > 0, \forall t \geq 0 \quad (17)$$

Proof: Subtracting (13)-(14) from (11)-(12), we can write the error dynamics of the observer as:

$$\dot{\tilde{v}}_i(t) = a_i(t) - L_v \text{sgn}\left(\tilde{v}_i(t)\right), \quad (18)$$

$$\begin{aligned} \dot{\tilde{a}}_i(t) = & -\left(k_p + \frac{k_d}{h}\right)\tilde{v}_i(t) - \left(k_p + \frac{1}{h}\right)(a_i(t) - \eta(t)) \\ & - \frac{1}{h}\dot{a}_{i-1}(t)\tilde{\tau}(t) - L_a\left(\eta(t) - \hat{a}_i(t)\right), \end{aligned} \quad (19)$$

where $\tilde{v}_i(t) = v_i(t) - \hat{v}_i(t)$, $\tilde{a}_i(t) = a_i(t) - \hat{a}_i(t)$ and $\tilde{\tau}(t) = \tau(t) - \hat{\tau}(t)$ are the estimation errors. We start with analyzing (18) by choosing the following Lyapunov function candidate: $V_v(t) = \frac{1}{2}\tilde{v}_i^2(t)$. Next, $\dot{V}_v(t)$ is given by:

$$\dot{V}_v(t) = \tilde{v}_i(t)\dot{\tilde{v}}_i(t) = \tilde{v}_i(t)a_i(t) - L_v\tilde{v}_i(t)\text{sgn}\left(\tilde{v}_i(t)\right), \quad (20)$$

Applying the inequality $AB \leq |A||B|$ on the first term of the right hand side of (20), we can write

$$\dot{V}_v(t) = |\tilde{v}_i(t)||a_i(t)| - L_v|\tilde{v}_i(t)| = |\tilde{v}_i(t)|(|a_i(t)| - L_v), \quad (21)$$

If the observer gain is such that $L_v > |a_i(t)| > 0, \forall t \geq 0$, then $\dot{V}_v(t) \leq 0$, and hence we can write:

$$\dot{V}_v(t) \leq -\alpha\sqrt{V_v(t)}, \quad (22)$$

where $\alpha = \min_{t \geq 0} \sqrt{2}(L_v - |a_i(t)|) \geq 0$. The solution of the differential inequality (22) is given by $V_v(t) \leq \left(-\frac{\alpha}{2}t + \sqrt{V_v(0)}\right)^2$. Therefore, we can conclude that $V_v(t) \rightarrow 0$ after some finite time $t_f < \frac{2}{\alpha}\sqrt{V_v(0)}$. After $t > t_f$, we have

$V_v(t) = 0$, $\dot{V}_v(t) = 0$, hence $\tilde{v}_i(t) = 0$, $\dot{\tilde{v}}_i(t) = 0$ [50]. Therefore, after $t > t_f$ we can re-write (18) as

$$0 = a_i(t) - \eta(t) \Rightarrow \eta(t) = a_i(t), \quad (23)$$

where $\eta(t)$ is called *equivalent output error injection* to maintain the sliding motion [50]. In practice, $\eta(t)$ can be extracted by passing the switching signal $L_v \text{sgn}(\tilde{v}_i(t))$ through a low-pass filter with unity steady-state gain. Next, we analyze the error dynamics (19) using the Lyapunov function candidate $V_a(t) = \frac{1}{2}\tilde{a}_i^2(t) + \frac{K}{2}\tilde{\tau}^2(t)$ where $K > 0$. Next, $\dot{V}_a(t)$ is given by:

$$\dot{V}_a(t) = \tilde{a}_i(t)\dot{\tilde{a}}_i(t) + K\tilde{\tau}(t)\dot{\tilde{\tau}}(t), \quad (24)$$

After $t > t_f$ we have $\eta(t) = a_i(t)$ and $\tilde{v}_i(t) = 0$. Hence, we can re-write (24) as

$$\dot{V}_a(t) = -\frac{1}{h}\tilde{a}_i(t)\dot{a}_{i-1}(t)\tilde{\tau}(t) - L_a\tilde{a}_i(t)(\tilde{a}_i(t)) + K\tilde{\tau}(t)\dot{\tilde{\tau}}(t), \quad (25)$$

Considering the fact τ is constant and hence $\dot{\tau} = 0$, we can re-write (25) as

$$\dot{V}_a(t) = -\frac{1}{h}\tilde{a}_i(t)\dot{a}_{i-1}(t)\tilde{\tau}(t) - L_a\tilde{a}_i(t)(\tilde{a}_i(t)) - K\tilde{\tau}(t)\dot{\tilde{\tau}}(t), \quad (26)$$

Applying the update law (15) and choosing $K = \frac{1}{L_b}$, (26) becomes

$$\dot{V}_a(t) = -L_a\tilde{a}_i^2(t) \leq 0, \quad (27)$$

This concludes the decaying behavior of $V_a(t)$ that is $V_a(t) \leq V(0)$. So, starting from any positive initial value of $V_a(0)$, $V_a(t) \rightarrow \gamma < \infty$ is bounded as $t \rightarrow \infty$. Hence, by recalling $V_a(t) = \frac{1}{2}\tilde{a}_i^2(t) + \frac{K}{2}\tilde{\tau}^2(t)$, we conclude that $\tilde{a}_i(t)$ and $\tilde{\tau}(t)$ are bounded as well.

Convergence of $\tilde{a}_i$: In this part, we prove $\tilde{a}_i(t) \rightarrow 0$ as $t \rightarrow \infty$. We derive the second derivative of Lyapunov candidate $V_a(t)$ with respect to time as

$$\ddot{V}_a(t) = -2L_a\tilde{a}_i(t)\dot{\tilde{a}}_i(t), \quad (28)$$

After $t > t_f$, replacing $\dot{\tilde{a}}_i(t)$ by

$$\dot{\tilde{a}}_i(t) = -\frac{1}{h}\dot{a}_{i-1}(t)\tilde{\tau} - L_a\tilde{a}_i(t), \quad (29)$$

we have

$$\ddot{V}_a(t) = 2\frac{L_a}{h}\tilde{a}_i(t)\dot{a}_{i-1}(t)\tilde{\tau} + 2L_a^2\tilde{a}_i^2(t), \quad (30)$$

As it mentioned earlier, $\tilde{a}_i(t)$ and $\tilde{\tau}$ are bounded. Furthermore, referring to *Assumption 4*, $\dot{a}_{i-1}$ is bounded. Hence, (30) shows the boundedness of $\ddot{V}_a(t) < \infty$ which equivalently verifies that $\dot{V}_a(t)$ is uniformly continuous. Now, applying Barbalat's lemma [51] on $\dot{V}_a(t)$ combined with the fact that $V_a(t)$ is bounded, we have $\dot{V}_a(t) \rightarrow 0$ as $t \rightarrow \infty$. Consequently, $\dot{V}_a(t) = -2L_a\tilde{a}_i^2(t) \rightarrow 0$ indicates that $\tilde{a}_i(t) \rightarrow 0$ as $t \rightarrow \infty$.

Convergence of $\tilde{\tau}$: In this part, we prove $\tilde{\tau} \rightarrow 0$ as $t \rightarrow \infty$. We know that

$$\int_0^\infty \dot{\tilde{a}}_i(t)dx = \tilde{a}_i(\infty) - \tilde{a}_i(0) = -\tilde{a}_i(0) < \infty, \quad (31)$$

Furthermore, $\dot{V}_a(t)$, $\tilde{a}_i(t)$ and $\tilde{\tau}$ are uniformly continuous. Also, referring to *Assumption 4*, second derivative of $a_{i-1}(t)$ exists and is finite which equivalently imply that, $\dot{a}_{i-1}(t)$ is uniformly continuous. Hence, using (29) we conclude $\dot{\tilde{a}}_i(t)$ is uniformly continuous. Therefore, by applying Barbalat's lemma [51], and considering the fact that $\tilde{a}_i(t)$ is bounded, we can conclude $\dot{\tilde{a}}_i(t) \rightarrow 0$ as $t \rightarrow \infty$. Next, considering (27), where $\dot{\tilde{a}}_i(t) \rightarrow 0$ and $\tilde{a}_i(t) \rightarrow 0$ as $t \rightarrow \infty$, it is clear that $\tilde{\tau} \rightarrow 0$ as $t \rightarrow \infty$.

Remark 4: The estimate of the delay parameter, $\hat{\tau}$, will be used to detect the occurrence of the DoS attack. Ideally, a DoS occurrence will be detected when $\hat{\tau} > 0$. Furthermore, the magnitude of $\hat{\tau}$ will serve as an estimate of the effect of DoS.

B. Robustness to Uncertainties

The effects of uncertainties have not been considered in the design of the detection observers. However, possible sources of uncertainty that affect the diagnostic scheme are: 1) unmodeled dynamics, 2) Radar measurement noise [52], [53] 3) inherent communication delay in a practical DSRC network, 4) Driver behavior which affects CACC gains e.g. K_p and K_d [54]. The presence of these uncertainties prohibits $\hat{\tau}$ from having the idealized property of being zero even in the absence of any DoS attack. One of the possible ways to deal with this is to use nonzero threshold set based on a realistic DSRC network behavior. The detection logic will be: DoS attack is detected when $\hat{\tau} > \delta$ and no DoS when $\hat{\tau} \leq \delta$ where δ is the threshold. The effect of the uncertainties on $\hat{\tau}$ will be suppressed below this threshold value.

Below are the guidelines for selection of constant threshold values for the evaluation of the residual:

Step 1: Collect $\hat{\tau}$ data under no DoS attack in normal DSRC network conditions from Monte-Carlo simulations or experimental studies.

Step 2: Plot the probability distribution of $\hat{\tau}$. An example probability distribution is shown in Fig.(3). In practice, this probability distribution will depend on uncertainties in the experimental data or of the Monte-Carlo study.

Step 3: Select a maximum allowable probability of false alarms.

Step 4: It can be seen from Fig.(3) that the probability of the false alarms can be computed by the following equation:

$$P_{FA} = \int_\delta^\infty p_0(x)dx, \quad (32)$$

where P_{FA} is the probability of DoS false alarm, δ is the selected threshold for DoS attack and $p_0(x)$ is the $\hat{\tau}$ probability distribution under no DoS attack in the DSRC network. The goal here is to select δ which will yield an acceptable P_{FA} .

IV. SIMULATION STUDIES

In this section, we present simulation studies to evaluate the effectiveness of the proposed scheme. To prepare a simulation test-bed for our case studies, we modeled dynamics of the platoon and the DSRC network via MATLAB(R2014b)/Simulink. MATLAB/Simulink allows us to simulate a real-time model

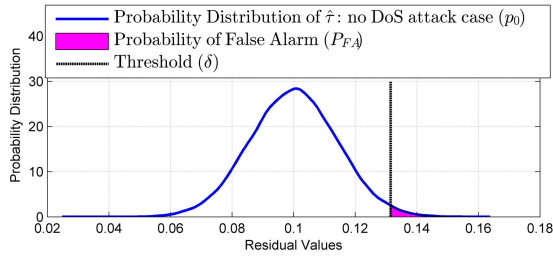


Fig. 3. Residual probability distribution under no attack condition.

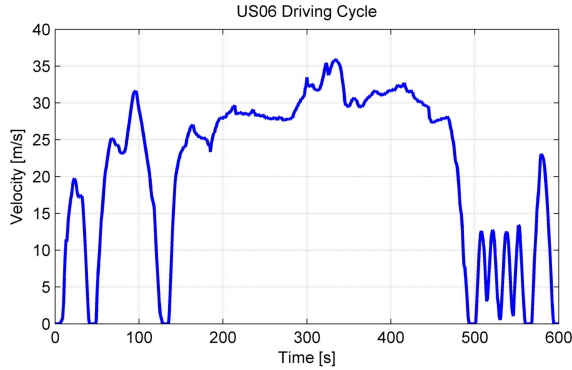


Fig. 4. US06 driving cycle profile.

of the system capturing both continuous nature of vehicle dynamics and discrete behavior of the vehicular communication network. In this simulation setup, the platoon consists of twelve vehicles under CACC strategy. The vehicles in the platoon exchange their safety related messages including acceleration information through DSRC network. Furthermore, we assume that the platoon follows a dynamic velocity profile, namely the US06 driving cycle shown in Fig. 4. In simulation, we repeat this driving cycle 13 times to create velocity trajectory followed by the leader vehicle. Model and control parameters of the platoon and CACC system are chosen as $h = 0.6$ s, $K_p = 0.7$, and $K_d = 1$ similar to the existing literature in this area [15], [25]. Since we have considered a random delay with mean value of 0.1 s as the nominal delay in the communication network, we have selected $h = 0.6$ s based upon the results in [21] and [22] to guarantee the stability of the string. To illustrate the results of the proposed algorithm, we particularly focus on the performance of the *vehicle 12* in the platoon. Since we focus on *vehicle 12*, the presented algorithm containing two observers implemented in *Vehicle 11*, as noted in Remark 3. Next, we present the following case studies.

Case 1: In this case study, we motivate the need for the DoS attack detection algorithm by illustrating the adverse effects of the attacks on the platoon. In Fig. 5, we show the relative distance (d_{12}), velocity (v_{12}) and acceleration (a_{12}) of *vehicle 12* in the presence of the DoS attacks. To simulate the effect of the DoS attacks, different magnitudes of delays are injected to the DSRC network. In the ideal case when there is no delay in DSRC network, i.e. $\tau = 0$ s, the relative distance of *vehicle 12* is maintained above the minimum safety

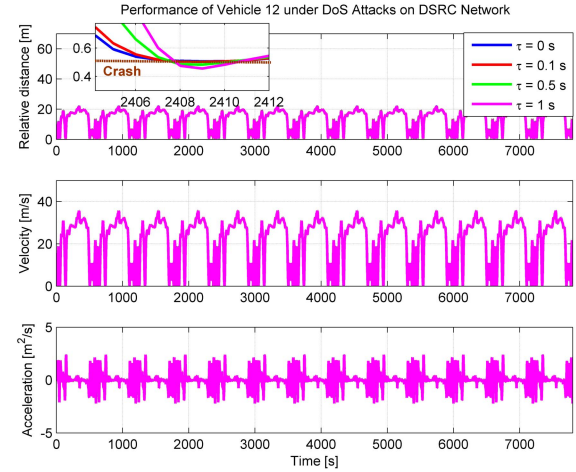


Fig. 5. Performance of the *vehicle 12* in the platoon under DoS Attacks in DSRC Network.

distance d_s for all time. However, the performance of *vehicle 12* degrades when we increase the magnitude of the DoS attacks, i.e. $\tau > 0.2$ s. In these scenarios, the minimum distance requirement is violated as shown in the top plot in Fig. 5. These violations represent crash scenarios in a platoon of self-driving vehicles. From this case study, we can conclude that DoS attack might lead to potentially dangerous situations. Hence, the need for DoS detection is evident for secure control of connected vehicles.

Case 2: In the second case study, we consider a more realistic scenario in the simulation. Ideal communication does not exist in practical applications, especially for DSRC [5]. Hence, we add a random non-zero mean delay with Gaussian distribution to the communication network. This random delay represents the inherent uncertainties in DSRC communication. The mean value and standard deviation of the network induced delay are chosen as $\mu_N = 0.1$ s and $\sigma_N = 0.03$ s, respectively. Next, we show the performance of the scheme in presence of this inherent communication delay. Note that, there is no DoS attack in this case study. The relative distance (d_{12}), velocity (v_{12}) and acceleration (a_{12}) corresponding to *vehicle 12* are shown in Fig. 6 under this scenario. We can see in the top plot of Fig. 6 that the CACC performs reasonably as the inter-vehicle distance between two cars is more than the assigned safety distance, $d_s = 0.5$ m. Regarding the detection scheme, observers are initialized with incorrect values of $\hat{a}_{12}(0) = 0$ and $\hat{\tau}(0) = 0$, respectively. The estimated value of acceleration in *vehicle 12* is shown in the first plot of Fig. 7. As can be seen in Fig. 7, the estimated value ($\hat{a}_{12}$) converges to its actual value (a_{12}). The estimation error is given in the bottom plot in Fig. 7. Furthermore, the detection algorithm is able to estimate the mean value of this network-induced delay as shown in the first plot of Fig. 8. We quantify the estimation performance in terms of convergence time. Referring to Fig. 8, the convergence time is within 100 seconds for the delay estimation. Importantly, the estimated delay is within the predefined threshold (δ) indicating no occurrence of DoS.

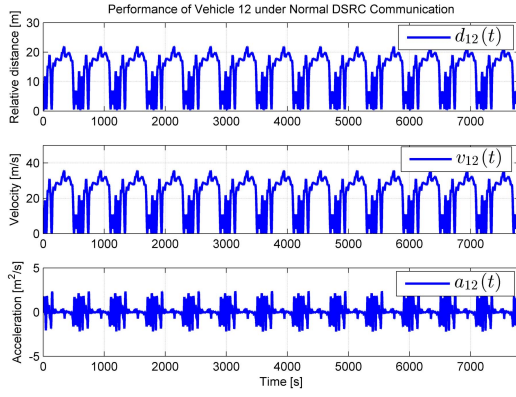
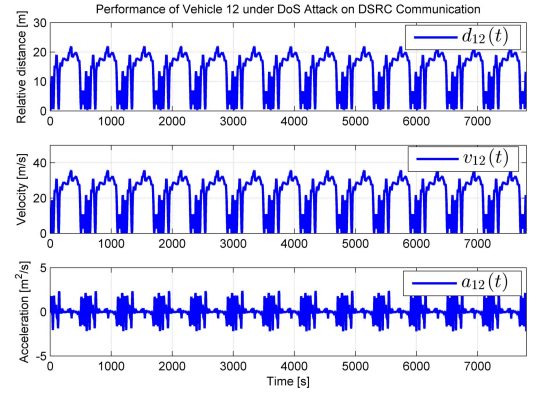
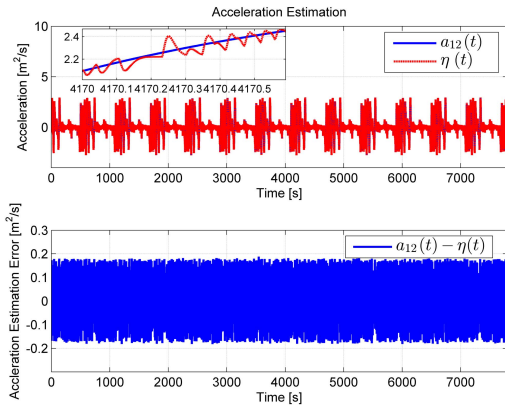
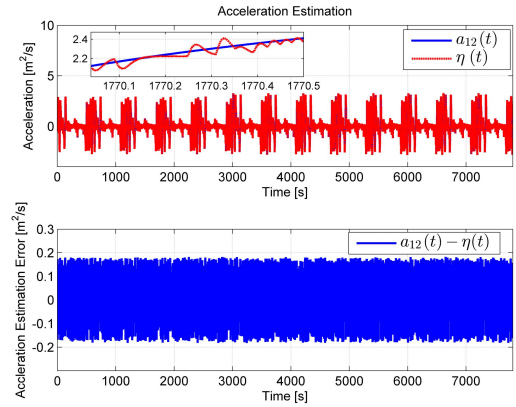
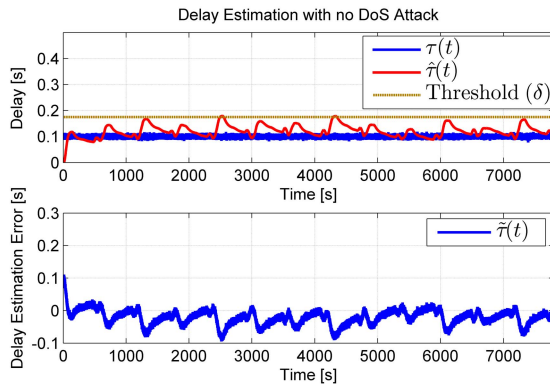

 Fig. 6. Performance of the *vehicle 12* in the platoon under normal DSRC network.

 Fig. 9. Performance of the *vehicle 12* in the platoon under DoS attack in DSRC network.

 Fig. 7. Acceleration estimation in *vehicle 12*. The variable a_{12} denotes actual value and η denotes estimated value.

 Fig. 10. Acceleration estimation in *vehicle 12*. The variable a_{12} denotes actual value and η denotes estimated value.


Fig. 8. Network induced delay and estimation of the delay in normal DSRC network.

Case 3: In this case, we test the effectiveness of the proposed approach under DoS attack. To emulate the DoS attack, a delay of $\tau = 0.5$ s is injected at $t = 3000$ s to the communication network apart from the aforementioned network induced delay. The induced delay takes certain time to reach a constant steady-state value as shown in Fig. 11. This is due to the assumption that it takes certain time for the attacker to jam the network. Figure 9 shows the relative distance d_{12} , velocity v_{12} and acceleration a_{12} of *vehicle 12*

in the presence of this DoS attack. Observers are initialized with incorrect values of $a_{12}(0) =$ and $\hat{\tau}(0) = 0$, respectively. The estimated acceleration of vehicle 12, $\hat{a}_{12}$ is shown in the top plot of Fig. 10 along with the actual value of a_{12} . The acceleration estimation error given by bottom plot of Fig. 10. These two figures illustrate that the estimated value $\hat{a}_{12}(t)$ converges to the actual value in finite time. Furthermore, the algorithm is also able to detect and estimate the DoS attack. The top plot in Fig. 11 shows the delay estimation performance. The estimate $\hat{\tau}$ closely tracks the true delay τ in the communication network. Hence, when the DoS attack level exceed the threshold, the estimated delay also surpasses the threshold very fast and indicate the occurrence of DoS attack in to the system. The attack is detected when the estimated delay ($\hat{\tau}$) crossed the threshold (δ) after the attack occurrence. The DoS estimation error is given in the bottom plot of Fig. 11. The steady-state delay estimation error lies within less than 10% of the original value.

Next, we illustrate the effectiveness of the proposed approach under several form of uncertainties. In all these following scenarios, a non-zero mean Gaussian delay with the mean value of $\mu_N = 0.1$ s and standard deviation of $\sigma_N = 0.02$ s is considered as the network induced delay. Furthermore, a constant delay of $\tau = 0.5$ s is added to represent the attack at $t = 3000$ s.

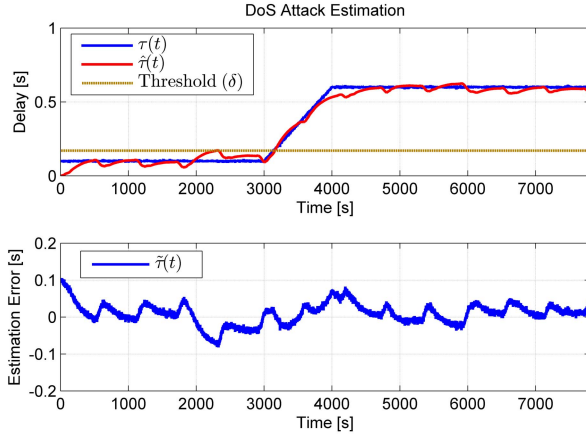
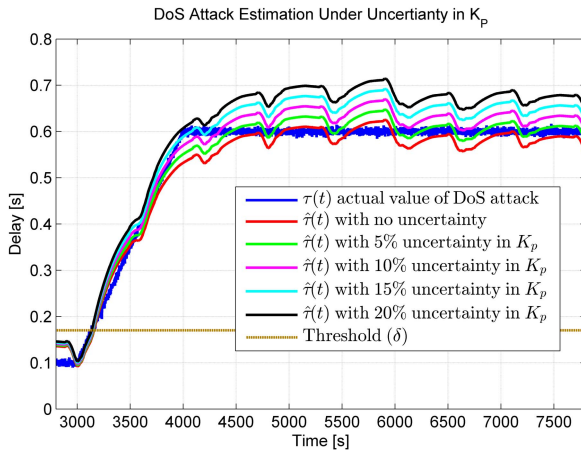


Fig. 11. Delay estimation under DoS attack.

Fig. 12. Delay estimation performance under different levels of uncertainties in the parameter K_p . The scenario is based on Case 4.

Case 4: In this case study we demonstrate the robustness of DoS detection algorithm to the uncertainty in the proportional gain K_p . Note that the observers are designed based on the nominal parameter value whereas the actual vehicle parameter is different than the nominal value. The nature of the uncertainty is an additive constant offset added to the nominal value of parameter K_p . We inject 5%, 10%, 15%, and 20% uncertainties to the K_p , i.e. $K_p = K_{p0} + \Delta K_p$ where K_{p0} is the nominal value and ΔK_p is the injected uncertainty. Figure 12 presents the estimated values of the delay under these uncertainties. As can be seen from Fig. 12, even in the presence of uncertainties, the proposed scheme can detect the DoS attack very fast when the DoS attack level surpasses the threshold. However, delay estimation suffers from these uncertainties leading to having approximately 10% error in estimation under 20% uncertainty in K_p .

Case 5: In this scenario, the robustness of the proposed scheme is evaluated under uncertain K_d . To study the effect of uncertainties, we inject 5%, 10%, 15%, and 20% uncertainties to the nominal value of K_d in the model. Same as before, the observers are designed based on the nominal parameter value. Figure 13 shows the delay estimation for different levels of uncertainties in K_d . As can be inferred from

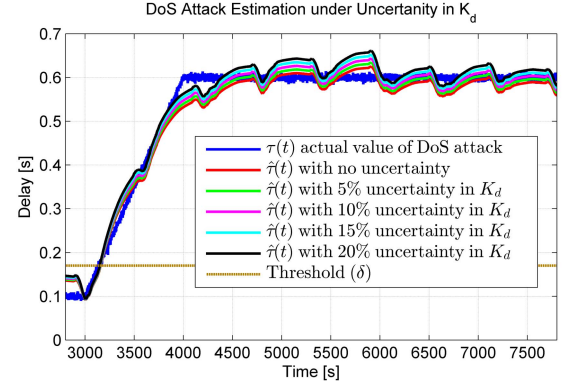
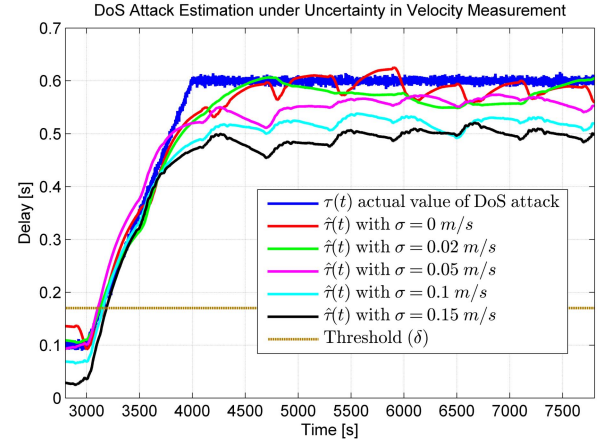
Fig. 13. Delay estimation performance under different levels of uncertainties in the parameter K_d . The scenario is based on Case 5.Fig. 14. Delay estimation performance under different levels of uncertainties in V_i measurement. The scenario is based on Case 6.

Fig. 13, the DoS detection algorithm can detect the DoS attack despite the uncertainties. However, the presence of uncertainties affects the accurate delay estimation. For example, there is 10% estimation error under 20% uncertain K_d .

Case 6: In this case study the effects of measurement noise is discussed. A zero mean Gaussian noise with standard deviation of σ is added to the velocity measurement. To illustrate the robustness of the proposed scheme under measurement uncertainty in V_i , different levels of noises as $\sigma = 0.02, 0.05, 0.1, 0.15$ are added to the velocity measurement. Figure 14 shows the delay estimation for different levels of measurement noises in V_i . It can be seen in Fig. 14 that the DoS detection algorithm detects the DoS attack in all cases. However, the presence of uncertainties affects the accurate delay estimation. For example, there is 20% estimation error under $\sigma = 0.15$.

V. CONCLUSION

In this work, we propose a real-time scheme for diagnosis of Denial of Service (DoS) cyber attack in connected vehicles. Under DoS, the attacker keeps the communication network busy by sending fake requests and hence the network is unable to respond to legitimate requests from the real users. Specifically, the proposed scheme can potentially (i) detect the occurrence of DoS, and (ii) estimate its effect on the connected vehicle system. We model the effect of the attack by

a time delay in the information processing via communication network. The main goal of the proposed scheme is to track this delay in information processing. The proposed scheme consists of a set of observers designed via sliding mode theory and adaptive observer theory. Simulation case studies are provided to verify the effectiveness of the proposed scheme. Furthermore, the robustness of the scheme is verified (i) under several forms of parametric uncertainties, and (iii) several measurement noise scenarios.

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